

Customer Churn Analysis Using SQL & Power BI

Project Overview

Customer churn is one of the most critical challenges faced by businesses. Retaining existing customers is often more cost-effective than acquiring new ones. This project analyzes customer churn data to identify factors influencing customer attrition and provides actionable business recommendations.

The project was completed using SQL for data cleaning, transformation, and business analysis, and Power BI for interactive dashboard development and visualization.

About the Dataset

The Customer Churn dataset contains information about bank customers and whether they have remained with the bank or exited (churned). The dataset is commonly used for customer retention analysis, predictive analytics, and business intelligence projects.

Dataset Overview

- Total Records: 10,000 Customers
- Total Columns: 14 Features
- Industry: Banking
- Target Variable: Exited (Customer Churn)

Dataset Features

Column Name	Description
Row Number	Unique row identifier
Customer Id	Unique customer ID
Surname	Customer surname
Credit Score	Customer credit score
Geography	Customer country (France, Germany, Spain)
Gender	Customer gender
Age	Customer age
Tenure	Number of years with the bank
Balance	Customer account balance
Num Of Products	Number of banking products owned
Has Cr Card	Whether customer owns a credit card (1 = Yes, 0 = No)
Is Active Member	Whether customer is an active member (1 = Yes, 0 = No)
Estimated Salary	Estimated annual salary
Exited	Churn status (1 = Customer Left, 0 = Customer Retained)

Business Problem

Customer churn directly impacts a bank's profitability and growth. The objective of this analysis is to identify the characteristics of customers who are most likely to leave the bank and provide actionable insights that can help improve customer retention and reduce churn rates.

Target Variable

Exited:

- 0 = Customer Retained
- 1 = Customer Churned

This column is used to measure customer attrition and analyze the factors influencing customer departure.

Objectives

- Analyze customer churn behavior.
- Identify customer segments with high churn risk.
- Discover patterns related to age, geography, products, balance, and credit score.
- Build an interactive dashboard for business stakeholders.
- Provide recommendations to improve customer retention.

Tools & Technologies

- ❖ SQL (MySQL)
- ❖ Power BI
- ❖ Excel / CSV Dataset
- ❖ GitHub

Project Workflow

Step 1: Database Creation

- Created database in MySQL.
- Created customer churn table structure.
- Imported CSV dataset into SQL.

Step 2: Data Cleaning

- Checked for null values.
- Verified duplicate records.
- Validated data types.
- Performed data quality checks.

Step 3: SQL Analysis

Conducted multiple business analyses including:

- **Total Customers**
- **Churned Customers**
- **Churn Rate Percentage**
- **Churn by Geography**
- **Churn by Gender**
- **Churn by Age Group**
- **Churn by Credit Score Category**
- **Churn by Number of Products**
- **Active vs Inactive Members**
- **Credit Card Ownership Analysis**
- **High Balance Churn Customers**
- **Customer Segmentation Analysis**

Step 4: Power BI Dashboard Development:

Created interactive dashboard with:

- KPI Cards
- Churn Rate Analysis
- Geography Analysis
- Age Group Analysis
- Product Analysis
- Credit Score Analysis
- Balance Analysis
- Customer Segmentation

Dashboard KPIs

- Total Customers
- Churned Customers
- Churn Rate %
- Average Balance
- Active Member %
- Average Credit Score

Key Business Insights

- ✚ Overall churn rate is 20.37%, meaning nearly 1 in 5 customers have left the bank.
- ✚ Germany has the highest customer churn, followed by France, indicating regional retention challenges.
- ✚ Customers aged 41–50 show the highest churn rate among all age groups.
- ✚ Inactive members are significantly more likely to churn than active members.
- ✚ Customers with 1 product churn more frequently than those with 2 products.
- ✚ Customers with poor credit scores have the highest churn rates.
- ✚ Several high-balance customers have churned, resulting in potential revenue loss.
- ✚ Female customers exhibit a slightly higher churn rate compared to male customers.

Business Recommendations

- ✓ Improve customer engagement programs for inactive customers.
- ✓ Implement cross-selling strategies to increase product adoption.
- ✓ Launch targeted retention campaigns for customers aged 41–50.
- ✓ Provide personalized offers for high-balance customers.
- ✓ Investigate churn drivers in Germany and France.
- ✓ Develop loyalty programs for customers with poor credit scores.
- ✓ Monitor churn trends regularly through dashboard reporting.

Project Outcome

This project demonstrates how SQL and Power BI can be used together to transform raw customer data into actionable business insights. The analysis helps identify churn drivers, improve customer retention strategies, and support data-driven decision-making.